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LCD monitors have been on the verge of mass usage for a long time now. But the fact that sales never took off despite them catching the fancy of computer users bears ample testimony to the fact that there are still some issues with the products that need to be sorted out—primarily, cost and value for money.

The good news is that prices have been dropping steadily over the past few months, to the point that many of us—even home users—can easily think of owning one. Nineteen-inch LCDs, unlike their CRT counterparts, remai beyond the reach of most of us, so we tested 15-inch and 17-inch LCD monitors—to be precise, 15 models from 15 brands in the 15-inch category, and 11 models from 10 brands in the 17-inch category.

Remember that unlike with CRTs, you don't get an inch or

two chopped off the viewable area—if it says 15 inches, almost all of it is viewable.

LCDs have suffered in the past in terms of acceptance because of their mediocre performance in the movies and games department, and also because of the restricted viewing angle. With better manufacturing processes, not only have prices come down, performance, too, in many areas has improved.

Apart from the fact that the technology is entirely different, LCD monitors differ in many non-intuitive ways from CRTs. For example, LCDs have a 'native resolution' at which they work best.

Remember, LCDs aren't the ideal solution for everyone out there—read on to find out whether to go the LCD way. We compared several LCDs based on aesthetic and technical aspects, as well as price—so you can make a sound decision on which one to buy.

Key Features

Here are some features to look at before you decide upon a model. Although our list is not exhaustive, it represents the most important things you need to know about and look out for.

Some things, such as the pixel pitch, are different from the CRT equivalent. Also, contrast ratios and such have different typical values in LCDs than CRTs.

Pixel Pitch

The pixel pitch is the distance between two white pixels, or two sub-pixels of the same colour. (A white pixel is made up of three sub-pixels—one red, one green and one blue.) The monitors we received in both the 15 and 17-inch categories had pixel pitches of 0.297 mm and 0.264 mm respectively—this is almost a universal standard. The individual size of each sub-pixel is, of course, smaller than the pixel pitch.

Native Resolution

In LCDs, the pixels are hard-etched on the panel, and therefore cannot display crisp images at any resolution other than the native resolution—unlike CRT monitors. The 15-inch and 17-inch models we received had native resolutions of 1024 x 768 and 1280 x 1024 pixels respectively—this, too, is almost universal.

If you change the resolution setting, you will see a badly focused image, and hard-to-read fonts with blurred edges.

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Luminance

Luminance and contrast ratio are of prime importance in LCDs. The luminance rating depends on the monitor's fluorescent lamp, built into the casing. The brighter the lamp, the better. The lamp rating is what determines the brightness level the monitor can achieve.

In the 15-inch category, the CMV CT-529A had the highest luminance rating: 450 cd/m^2 (candelas per square metre). This is the vendor-specified rating, and, if correct, it means that the LCD can look bright even in a fully-lit room.

The Acer AL1512 claims a luminance rating of 350 cd/m^2, which is higher compared to the rest of the bunch which maxed out at 250 cd/m^2. The Philips 150s had the lowest luminance rating of 200 cd/m^2, which means you need to make sure light doesn't hit the screen directly.

In the 17-inch monitor category again, the CMV CT-712A had a 400 cd/m^2 luminance rating, followed by the ViewSonic and the Acer at 380 cd/m^2 and 370 cd/m^2 respectively. The CMV's better lamp makes it easy for the user to tweak the brightness of the monitor according to the ambient light.

Contrast Ratio

The contrast ratio is the ratio of the brightness of pure white versus pitch black that the monitor displays. A higher contrast ratio means the user can set the monitor to display images with better visual depth. Also, a monitor with a higher contrast ratio helps the user fine-tune the panel to suit different lighting conditions.

Although LCD monitors do come with an 'Auto' setting feature that takes care of the brightness, contrast and geometry, this feature is usually not perfect,

Viewsonic VE510b

Umax Maxvision V5

and the user almost always has to do some tweaking. A contrast ratio of 120:1 is enough to display most colours. However, an even higher contrast ratio allows the panel to display more greyscales. This means you'll be able to see more shades of colour on an LCD with a higher contrast ratio.

Among the 15-inchers, the CMV and the HCL claimed the highest contrast ratio of 500:1, which is impressive, and which means the user has a wide range